

Azure Databricks Architecture — Knowledge Check

20 questions · Understand Azure Databricks architecture (Microsoft Learn). Answer key starts on a new page.

1. What is the top-level construct used to manage Azure Databricks across an entire organization — including identity/access, workspace creation, and billing?

- A. Workspace
- B. Metastore
- C. Account
- D. Catalog

2. Which Unity Catalog object is region-specific and serves as the top-level container for metadata (tables, views, volumes, models, functions)?

- A. Catalog
- B. Metastore
- C. Workspace
- D. Schema

3. What is the 3-level namespace format Unity Catalog uses to organize data objects?

4. After a workspace is enabled for Unity Catalog, how does the legacy Hive metastore become accessible?

- A. It's automatically deleted
- B. It appears as a catalog named hive_metastore
- C. It becomes the default schema
- D. It merges directly into Unity Catalog tables

5. True or False: The control plane processes and stores your actual data.

6. Where do serverless compute resources actually run?

- A. In your own Azure subscription
- B. In a serverless compute plane inside the Azure Databricks account
- C. On-premises servers
- D. Inside the control plane

7. What is another name for classic workspaces, as shown in the Azure portal?

- A. Serverless workspaces
- B. Hybrid workspaces
- C. Unity workspaces
- D. Managed workspaces

8. Name the two categories of data that live in workspace storage, regardless of workspace type.

9. Which of the following is an example of workspace SYSTEM data (not file system data)?

- A. Notebooks
- B. SQL query results and cluster logs
- C. Repos (Git-linked folders)

D. Libraries (.whl, .jar files)

10. True or False: DBFS root and DBFS mounts are the current best-practice way to store and access data in Azure Databricks.

11. Per current Unity Catalog behavior, what happens to a managed table's underlying files after you drop it?

- A. Deleted immediately with no recovery option
- B. Retained for a 7-day recovery window, then purged within 48 hours
- C. Retained indefinitely by default
- D. Immediately moved to external storage

12. At which level does Databricks recommend configuring a Unity Catalog managed storage location for most organizations?

- A. Metastore level
- B. Catalog level
- C. Schema level only
- D. Account level

13. List Unity Catalog's storage location resolution order when creating a managed table (most specific to most general).

14. What does Unity Catalog automatically append to a specified storage root to guarantee each catalog/schema gets a unique path?

- A. A timestamp
- B. The workspace name
- C. Hashed subdirectories (e.g., __unitystorage/catalogs/)
- D. A random file name

15. What two components make up a Unity Catalog external location?

- A. A workspace ID and a region
- B. A cloud storage path and a storage credential
- C. A catalog and a schema
- D. A metastore and an account

16. Which authentication mechanism does Databricks recommend for storage credentials, over service principals?

- A. SAS tokens
- B. Azure managed identities
- C. Shared access keys
- D. Basic username/password auth

17. For external tables/volumes, who manages the underlying data files, and where does the data remain?

18. What does "workspace binding" do for an external location or storage credential?

- A. Grants access to all workspaces automatically
- B. Restricts usage to specific, designated workspaces regardless of a user's Unity Catalog privileges
- C. Permanently deletes it if unused
- D. Converts it into a managed storage location

19. Which internal Azure Databricks features store their operational data in default storage, regardless of workspace type (classic or serverless)?

- A. Data Classification, Anomaly Detection, Clean Rooms, Knowledge Assistant
- B. Only Unity Catalog metastores
- C. Only DBFS
- D. Only external tables

20. Which statement about default storage is TRUE?

- A. Classic compute clusters can freely read/write default storage catalogs
- B. New catalogs can only be created on default storage in serverless workspaces
- C. Default storage requires you to configure your own cloud storage account first
- D. BI tools like Power BI cannot connect to default storage at all

Answer Key

1. C. Account

The account manages identity/access, creates workspaces, attaches metastores, and oversees billing and policies org-wide.

2. B. Metastore

A metastore is region-specific and registers metadata plus permissions; workspaces attached to it share a unified data view.

3. ..

This namespace lets Unity Catalog operate at the account level rather than per-workspace like the legacy Hive metastore.

4. B. It appears as a catalog named hive_metastore

You can query it as hive_metastore..., but Databricks recommends migrating fully to Unity Catalog.

5. False

The control plane hosts the web app and handles orchestration, job scheduling, and cluster management — it never processes your data. Data processing happens in the compute plane.

6. B. In a serverless compute plane inside the Azure Databricks account

This is the key distinction from classic compute, which runs inside your own subscription's virtual network.

7. B. Hybrid workspaces

Classic Azure Databricks workspaces are labeled "Hybrid workspaces" in the Azure portal UI.

8. Workspace file system data and workspace system data

File system data = notebooks, queries, repos, libraries. System data = query results, job results, notebook revisions, query plans, cluster logs.

9. B. SQL query results and cluster logs

System data is generated internally by Azure Databricks features, distinct from user-created file system assets.

10. False

DBFS root/mounts are legacy and deprecated. Use Unity Catalog-managed tables and volumes instead for governance and security.

11. B. Retained for a 7-day recovery window, then purged within 48 hours

You can UNDROP a table within the 7-day window. After that, Unity Catalog permanently purges the data files within 48 hours.

12. B. Catalog level

Catalog-level storage gives a clear governance boundary and logical data isolation; it's the recommended default approach.

13. Schema → Catalog → Metastore

UC checks the schema first, falls back to the catalog, then the metastore. If none are configured, managed tables/volumes can't be created.

14. C. Hashed subdirectories (e.g., __unitystorage/catalogs/)

This lets multiple catalogs/schemas safely share the same base storage root without path collisions.

15. B. A cloud storage path and a storage credential

The storage credential supplies authentication; the external location ties that credential to a specific storage path.

16. B. Azure managed identities

Managed identities need no secret rotation and can access storage accounts protected by network firewalls — service principals require manual secret management.

17. You (the customer) manage the data files; the data remains in its original storage location

This contrasts with managed storage locations, where Unity Catalog manages the full data lifecycle.

18. B. Restricts usage to specific, designated workspaces regardless of a user's Unity Catalog privileges

A common pattern: bind production credentials/locations to production workspaces only, keeping them out of reach from dev environments.

19. A. Data Classification, Anomaly Detection, Clean Rooms, Knowledge Assistant

These features always use default storage for their operational data, even in classic workspaces.

20. B. New catalogs can only be created on default storage in serverless workspaces

Classic compute cannot interact with default storage at all — even classic workspaces need serverless compute to read a default-storage catalog. BI tools CAN connect via ODBC/JDBC.